

The Horizon Map: Independent Mathematical Audit

Cutoffs, mass, spectral colors, and the exact growth coefficient

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Calculations and audit prepared with ChatGPT

October 6, 2026

Scope. This audit checks the pasted Claude report and its four-panel figure against the displayed definitions. It independently derives the identities and reproduces selected numerical values. The external files `THE_HORIZON_MAP.md` and its generating script were not inspected; their contents and repository status are not certified here.

Verdict. The arithmetic field representation and the distinction between integer and prime cutoffs are sound. Several interpretations require correction: both finite spectra are tempered; varying only the field mass cannot change finite versus infinite limiting energy; and the prime-product energy has leading logarithmic growth coefficient 4, whereas its norm has coefficient 2.

Working premise retained. The light-plane programme's growth law is

$$\text{for every } \varepsilon > 0, \quad R(N) \leq C_\varepsilon N^\varepsilon \quad (N \geq 1). \quad (\text{G})$$

It is an explicit premise throughout. The checks below distinguish consequences of (G), identities that need no growth premise, and finite numerical observations.

1. The common field and its two cutoffs

Put $G_v(u) = e^{-|u-v|/2}$ and

$$H_0 = -\frac{d^2}{du^2} + \frac{1}{4}, \quad \|f\|_{\mathcal{E}}^2 = \int_{\mathbb{R}} \left(|f'|^2 + \frac{1}{4}|f|^2 \right) du.$$

Then $H_0 G_v = \delta_v$ and $\langle G_v, G_w \rangle_{\mathcal{E}} = e^{-|v-w|/2}$. Consequently,

$$\phi_N = \sum_{n \leq N} \frac{\mu(n)}{\sqrt{n}} G_{\log n}, \quad R(N) = \|\phi_N\|_{\mathcal{E}}^2 = \sum_{a, b \leq N} \frac{\mu(a)\mu(b)}{\max(a, b)}.$$

If $q_P = \prod_{p \leq P} p$, the untruncated prime-cutoff field is instead

$$\Phi_P = \sum_{d|q_P} \frac{\mu(d)}{\sqrt{d}} G_{\log d}, \quad E_P = \|\Phi_P\|_{\mathcal{E}}^2.$$

The prime cutoff retains all squarefree products of its primes, up to q_P . It does not retain just the integers $n \leq P$. This difference explains why the two plots measure different families of sources.

2. Exact spectral identities and what is conserved

Define

$$D_N(s) = \sum_{n \leq N} \mu(n) n^{-s}, \quad F_P(s) = \prod_{p \leq P} (1 - p^{-s}) = \sum_{d|q_P} \mu(d) d^{-s}.$$

With Fourier transform $\widehat{f}(k) = \int f(u) e^{-iku} du$,

$$R(N) = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{|D_N(\frac{1}{2} + ik)|^2}{k^2 + \frac{1}{4}} dk,$$

$$E_P = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{|F_P(\frac{1}{2} + ik)|^2}{k^2 + \frac{1}{4}} dk.$$

These are weighted Plancherel identities for the same positive field metric. They prove positivity for both cutoff choices.

Finite spectra are tempered on both sides. Each displayed spectrum is a finite sum of real-frequency exponentials. For all real k ,

$$|D_N(\frac{1}{2} + ik)| \leq \sum_{n \leq N} n^{-1/2}, \quad |F_P(\frac{1}{2} + ik)| \leq \prod_{p \leq P} (1 + p^{-1/2}).$$

Thus both are bounded functions, and both define tempered distributions. Large variation as P changes is not non-temperedness in k . A statement about an infinite-cutoff distribution requires a specified topology and limiting object.

The product and its logarithm must be distinguished. For finite P and $\Re s > 0$, the convergent logarithmic series gives

$$\log F_P(s) = - \sum_{p \leq P} \sum_{r \geq 1} \frac{p^{-rs}}{r}.$$

In the absolutely convergent half-plane $\Re s > 1$, both cutoff families tend to $1/\zeta(s)$; the logarithmic series tends to $-\log \zeta(s)$ there. Neither an unregularized critical-line limit nor convergence at a zero follows from these finite identities. In particular, a finite D_N has no poles.

The prime update is exact for the full prime cutoff. Write $T_a f(u) = f(u - a)$. Adding a new prime q gives

$$\Phi_{P \cup \{q\}} = \Phi_P - q^{-1/2} T_{\log q} \Phi_P,$$

where $P \cup \{q\}$ denotes the enlarged prime set. Hence

$$E_{P \cup \{q\}} = (1 + q^{-1}) E_P - 2q^{-1/2} C_q, \quad C_q = \Re \langle \Phi_P, T_{\log q} \Phi_P \rangle_{\mathcal{E}}.$$

Translation preserves the energy of each copy. The coefficient $1 + 1/q$ adds their separate squared norms; the interference controls the energy of their sum. If $C_q < 0$, the cross contribution $-2q^{-1/2} C_q$ is *positive*. No sign theorem for every subsequent prime follows from a finite sign table.

3. What changing the mass actually means

For $m > 0$, set $H_m = -\partial_u^2 + m^2$. Its Green function is $G_m(u) = e^{-m|u|}/(2m)$. The original operator has *mass* $m = 1/2$ and *mass squared* $m^2 = 1/4$.

Fixed source, changed operator. Retain $J_N = \sum_{n \leq N} \mu(n)n^{-1/2}\delta_{\log n}$ and define its field energy under H_m by

$$E_m(J_N) = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{|D_N(\frac{1}{2} + ik)|^2}{k^2 + m^2} dk.$$

For every N ,

$$\min\{1, (4m^2)^{-1}\}R(N) \leq E_m(J_N) \leq \max\{1, (4m^2)^{-1}\}R(N).$$

Indeed, the ratio of the new weight to the original weight is $(k^2 + 1/4)/(k^2 + m^2)$, which lies between these two constants. They are independent of N .

Therefore a fixed source cannot have infinite limiting energy at mass $1/2$ and bounded limiting energy at every larger fixed mass. All positive masses give equivalent norms on this fixed family. This is a normalization issue in the proposed mass formulation of (G), not a new arithmetic obstruction.

Matched source and mass. To implement the intended weighted growth test, vary the source at the same time:

$$J_{m,N} = \sum_{n \leq N} \mu(n)n^{-m}\delta_{\log n}.$$

Its energy has the exact form

$$\begin{aligned} E_m(J_{m,N}) &= \frac{1}{2m} \sum_{a,b \leq N} \frac{\mu(a)\mu(b)}{\max(a,b)^{2m}} \\ &= \int_1^N \frac{M(x)^2}{x^{2m+1}} dx + \frac{M(N)^2}{2mN^{2m}}. \end{aligned}$$

This follows from $\max(a,b)^{-2m}/(2m) = \int_{\max(a,b)}^{\infty} x^{-2m-1} dx$. Under (G), $M(N)^2 \leq NR(N) \ll_{\varepsilon} N^{1+\varepsilon}$; choosing $\varepsilon < 2m - 1$ proves finite limiting energy whenever $m > 1/2$. This is the consistent family for the asserted critical mass threshold.

At $m = 1/2$, the historical integral cannot be finite. For completeness, if $I = \int_1^{\infty} M(x)^2 x^{-2} dx < \infty$, its Mellin transform would be holomorphic on $\Re s > 1/2$, equal $1/(s\zeta(s))$ there by continuation, and bounded by $\sqrt{I/(2\Re s - 1)}$. Approaching any known critical-line zero would give a pole of order at least one, contradicting this square-root bound. Thus critical energy diverges; this statement does not impose a logarithmic rate of divergence.

4. The prime-cutoff growth coefficient

Proposition 1. *For the prime-cutoff energy defined in Section 2,*

$$\boxed{\log E_P \sim \frac{4\sqrt{P}}{\log P}}, \quad \boxed{\log \|\Phi_P\|_{\mathcal{E}} \sim \frac{2\sqrt{P}}{\log P}} \quad (P \rightarrow \infty).$$

This conclusion uses the prime number theorem and does not use (G).

Proof. Let $A(P) = \sum_{p \leq P} p^{-1/2}$. Partial summation of the prime number theorem gives $A(P) \sim 2\sqrt{P}/\log P$. Also

$$\log |F_P(\tfrac{1}{2} + ik)|^2 = -2 \sum_{p \leq P} p^{-1/2} \cos(k \log p) + O(\log P)$$

uniformly in real k : the remainder of $\log(1 - z) = -z + O(|z|^2)$ is uniform for $|z| \leq 1/\sqrt{2}$, and $\sum_{p \leq P} 1/p = O(\log P)$ suffices. The error is $o(A(P))$.

For the upper bound, $|F_P(\tfrac{1}{2} + ik)| \leq e^{A(P)}$ and

$$\frac{1}{2\pi} \int_{\mathbb{R}} \frac{dk}{k^2 + 1/4} = 1.$$

Thus $E_P \leq e^{2A(P)}$.

For the lower bound fix $0 < \delta < 1/10$ and restrict k to

$$I_P = \left[\frac{\pi - \delta}{\log P}, \frac{\pi + \delta}{\log P} \right].$$

For $P^{1-\delta} \leq p \leq P$ and $k \in I_P$,

$$|k \log p - \pi| \leq (\pi + 1)\delta, \quad -\cos(k \log p) \geq \cos((\pi + 1)\delta).$$

The remaining primes have weight $A(P^{1-\delta}) = o(A(P))$. Hence, uniformly on this interval,

$$\log |F_P(\tfrac{1}{2} + ik)|^2 \geq \{2 \cos((\pi + 1)\delta) + o(1)\} A(P).$$

On I_P the propagator weight is bounded below by a fixed positive constant for sufficiently large P . The interval has length $2\delta/\log P$, whose logarithm is $o(A(P))$. Integrating over I_P proves

$$\liminf_{P \rightarrow \infty} \frac{\log E_P}{A(P)} \geq 2 \cos((\pi + 1)\delta).$$

Let $\delta \downarrow 0$ and use the upper bound to obtain $\log E_P \sim 2A(P) \sim 4\sqrt{P}/\log P$. Taking the square root of the energy gives the norm formula. $\square$

Thus the coefficient 2 is appropriate for the *norm*, and 4 for the *energy* plotted as a squared norm. Finite ranges need not yet display this asymptotic coefficient accurately.

There is also a useful scale comparison: $\log q_P \sim P$, so the same theorem gives $E_P = q_P^{o(1)}$. The growth is subpower when measured against the largest integer represented in the prime cutoff. This does not bound every integer prefix, but it prevents confusing fast growth in P with fast growth in q_P .

5. Independent checks of the displayed numbers

The accompanying `checks.py` and `checks.json` record reproducible double-precision calculations. They test finite claims only.

For $P = 10^4$, evaluating the spectrum on $0 \leq k \leq 60$ in steps of 0.001 and refining the largest sampled peak gives:

Measurement	Value
Minimum sampled $\log_{10} F_P ^2$	-26.845011
Refined maximum $\log_{10} F_P ^2$	15.794478
Location of refined maximum	$k = 0.416049$
Observed range in decades	42.639489

This supports the reported roughly 43-decade swing on the plotted interval. The numerical search is not a certified global extremum computation.

The stated half-unit accuracy of the dashed first-prime-sum approximation fails at $k = 0$ if it means a uniform error on the plotted interval. In natural logarithms of the *squared* magnitude, the exact discrepancy there is

$$2 \sum_{p \leq 10^4} \left[\log(1 - p^{-1/2}) + p^{-1/2} \right] = -3.525536966 \dots$$

For the logarithm of the magnitude rather than its square, the discrepancy is $-1.762768483 \dots$, still larger than 0.5 in absolute value. A claim restricted to a smaller frequency range would need that range stated.

Integer energies, with the terminal contribution retained, are:

N	$R(N)$	$\sum_{n \leq N} \mu(n)^2/n$	Ratio
10^2	1.361579213	3.835567134	0.354988
10^3	1.459409418	5.242393129	0.278386
10^4	1.582361327	6.643413164	0.238185
10^5	1.621566526	8.042911619	0.201614
10^6	1.708249134	9.442716145	0.180907
10^7	1.836533426	10.842523422	0.169382

The reported endpoint fractions 0.28 and 0.17 are reproduced. The slope 0.0278 is not uniquely specified without the original sampling and fitting rule. Our least-squares fit of $R(N)$ against $\log N$ on 301 rounded, logarithmically spaced points from 10^4 to 10^7 gives 0.029306; the endpoint secant gives 0.036795. Neither is an all-scale growth law.

For full prime cutoffs, $E_2 = 1/2$, $E_3 = 2/3$, $E_5 = 13/15$, and $E_7 = 116/105$. The correlation is $C_2 = 1/\sqrt{2}$, $C_3 = 0$, and negative for each tested new prime from 5 through 37. This confirms the initial sign change, not a universal subsequent sign claim.

6. Mean density, zero constants, and the pole approximation

For every fixed N , expanding the square and averaging its distinct frequencies proves

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |D_N(\tfrac{1}{2} + ik)|^2 dk = \sum_{n \leq N} \frac{\mu(n)^2}{n} = \frac{6}{\pi^2} \log N + O(1).$$

The last equality follows from squarefree counting and partial summation. This is an exact long-frequency mean identity followed by an asymptotic. Ordinary unweighted $L^2(\mathbb{R})$ Plancherel is not the relevant claim, since these nonzero finite exponential sums are not in $L^2(\mathbb{R})$. The field energy uses the weighted Plancherel identity in Section 2. The diagonal is a useful reference level, but has not been identified as a physical conserved incoming flux.

The conditional zero constant. A precise result of Ng states that, assuming RH and $J_{-1}(T) = \sum_{0 < \gamma \leq T} |\zeta'(1/2 + i\gamma)|^{-2} \ll T$,

$$\int_1^X \frac{M(x)^2}{x^2} dx \sim \beta \log X, \quad \beta = 2 \sum_{\gamma > 0} \frac{1}{|(1/2 + i\gamma)\zeta'(1/2 + i\gamma)|^2}.$$

The derivative-moment hypothesis includes simplicity of the zeros. It is additional information beyond (G). This theorem concerns the historical integral; the complete finite energy also contains $M(N)^2/N$, which must be accounted for when comparing fitted slopes. The reported decimal 0.0288 cannot be certified without its zero list, truncation, and tail control.

The pole envelope is a useful leading model. For fixed k , prime number theorem partial summation yields the leading endpoint contribution

$$\sum_{p \leq P} p^{-1/2 - ik} \sim \frac{P^{1/2 - ik}}{(1/2 - ik) \log P}.$$

Accordingly, the leading oscillatory term in $\log |F_P|^2$ has envelope size

$$\frac{2\sqrt{P}}{\sqrt{k^2 + 1/4} \log P}.$$

This supports an explanation through the density of primes associated with the zeta pole. It does not by itself prove that a zero becomes visible exactly at $k = \sqrt{P}/\log P$. Such a visibility claim needs a defined contrast criterion and estimates uniform in the growing frequency range. The dashed curve computed from the *actual prime sum* also contains arithmetic fluctuations; it is not literally the isolated pole contribution.

Hamiltonian spectrum. $H_{\text{dil}} = -i\partial_u$ and $H_0 = H_{\text{dil}}^2 + 1/4$ have spectral value $k^2 + 1/4 = s(1 - s)$ at $s = 1/2 + ik$. This equality is correct. Their free-field spectrum is continuous; the finite arithmetic source determines which colors are excited. The finite plots do not convert these colors into normalizable eigenstates at the zeta zeros.

7. A common map that keeps the comparison precise

The most direct joint object is

$$\Phi_{P,N}(u) = \sum_{\substack{d|q_P \\ d \leq N}} \frac{\mu(d)}{\sqrt{d}} G_{\log d}(u).$$

It has two exact specializations:

$$P \geq N \implies \Phi_{P,N} = \phi_N, \quad N \geq q_P \implies \Phi_{P,N} = \Phi_P.$$

Adding a new prime q preserves the exact smaller horizon:

$$\Phi_{P \cup \{q\}, N} = \Phi_{P,N} - q^{-1/2} T_{\log q} \Phi_{P, \lfloor N/q \rfloor}.$$

This is where the integer cutoff changes the update. Replacing the second field by $\Phi_{P,N}$ without removing the integer cutoff changes the source. Its energy expansion contains both squared norms and their full mixed term.

Object or claim	Precise status
Green kernel and energy	The same kernel and quadratic metric apply to both cutoffs. Their numerical energy values generally differ.
Translation conservation	T_a is unitary in the field energy metric; adding a translated copy is a different operation.
Prime construction	The primes generate a free <i>commutative</i> monoid. Sampling does not establish an isomorphism from the continuous group $\mathbb{R}$.
Commutator	On finite-support arithmetic states, $[H_{\text{arith}}, S_p] = (\log p)S_p$, with $H_{\text{arith}} n\rangle = \log n n\rangle$. This covariance is exact; it is not the literal same commutator as $[u, -i\partial_u] = i$.
Equipartition	For any finite source of this form, the inward and outward channels each contribute half of the total field energy.
Positivity	The propagator remains positive for either cutoff. Positivity alone supplies no upper bound on source-dependent energy.

Corrected interpretation. One positive field geometry accommodates two different truncations of the same multiplicative coefficient rule. Translation preserves its norm, while assembling signed copies changes the energy through interference. The integer growth law is retained as (G). The full prime-cutoff energy has the unconditional asymptotic proved in Section 4. The two-cutoff field above identifies exactly which source terms are retained in each comparison. Any proposed order-of-limits statement must specify its observable, normalization, and mode of convergence.

References. NIST DLMF, §27.12 (prime number theorem), <https://dlmf.nist.gov/27.12>; §27.4 (Euler products), <https://dlmf.nist.gov/27.4>. Nathan Ng, *The distribution of the summatory function of the Möbius function*, Theorem 3, manuscript pp. 8–9, <https://www.cs.uleth.ca/~nathanng/RESEARCH/mobius2b.pdf>. Richard Melrose, *Tempered distributions and the Fourier transform*, <https://math.mit.edu/~rbm/iml/Chapter1.pdf>. The preceding project note, *The Light Plane: An Exact Horizon Map* (October 5, 2026), supplies the established notation and framework premise.

8. Follow-up: phase readout and the omitted tail

Claude's follow-up identifies the source-phase parameter with the center of the color readout. This identification is exact. Define

$$A_N(k) = |D_N(\tfrac{1}{2} + ik)|^2, \quad L(t) = \frac{1}{2\pi(t^2 + 1/4)}.$$

Then $\int_{\mathbb{R}} L(t) dt = 1$ and

$$R_N(\tau) = \int_{\mathbb{R}} A_N(k) L(k - \tau) dk.$$

Thus τ selects the center of a normalized Lorentzian window on the color coordinate k . Its half-width at half-maximum is $1/2$; its full width is 1. The readout includes the whole spectrum with these weights. Expanding the finite square also gives the exact long-phase mean

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T R_N(\tau) d\tau = \sum_{n \leq N} \frac{\mu(n)^2}{n}.$$

The off-diagonal phase terms have zero mean.

Independent calculations, using the historical sum to evaluate phase energy, give:

N	$R_N(0)$	$R_N(14.1347)$	Sine-resolution error
1000	1.459409418	33.547248500	$< 10^{-15}$
10000	1.582361327	49.603937401	$< 10^{-15}$

The maximum inversion errors in this implementation are $2.42 \cdot 10^{-12}$ and $1.89 \cdot 10^{-10}$, respectively. The phase bound holds at the six tested phases at both cutoffs; its analytic proof already covers all real phases. Reported percentages of the (G) mode allowance require the chosen $\varepsilon, C_\varepsilon, N, J$ and were not independently reproduced.

The tail-dominance statement can be proved precisely. Let $n_P = \lfloor P \rfloor$ and

$$\mathcal{T}_P = \Phi_P - \phi_{n_P} = \sum_{\substack{d|q_P \\ d > P}} \frac{\mu(d)}{\sqrt{d}} G_{\log d}.$$

The common field metric retains every mixed term. The elementary estimate

$$R(n_P) \leq \sum_{a,b \leq n_P} \frac{1}{\max(a,b)} = 2n_P - \sum_{j \leq n_P} \frac{1}{j} \leq 2P$$

and the norm triangle inequality imply

$$\left| \frac{\|\mathcal{T}_P\|_{\mathcal{E}}}{\sqrt{E_P}} - 1 \right| \leq \sqrt{\frac{R(n_P)}{E_P}} \leq \sqrt{\frac{2P}{E_P}} \rightarrow 0.$$

Here Section 4 supplies $\log E_P \sim 4\sqrt{P}/\log P$, which grows faster than $\log P$. Therefore

$$\|\mathcal{T}_P\|_{\mathcal{E}}^2 \sim E_P.$$

This proves dominance in the field energy norm, without (G). It does not assert pointwise dominance at every color. It gives an exact meaning to the follow-up's main cutoff observation, while retaining the corrected energy coefficient and the full inherited cutoff in each prime update.